

Bachelor's Thesis Proposal

A Domain-Specific Language for Privacy-Aware Business Processes

Assessor: Dr. Hoang Nguyen
Co-assessor: To be decided

1 Motivation

Organizations describe how they operate in terms of *business processes*. The Business Process Model and Notation (BPMN) [2] is the modeling standard for business analysts and developers. It is supported by a large ecosystem of tools for drawing, validating, simulating, and even executing process models on workflow engines.

With the increasing interest in protecting users' personal data in software systems, a natural extension of BPMN is to integrate the declaration of privacy concepts into the processes. Ideally, the software designer should be able to define a business process model in which each activity or process is associated with a declared purpose. For example, a process for sending mass marketing emails to customers can be associated with a **MARKETING** purpose. This idea has been investigated and reported in the work of Basin et al. [1].

2 Objective

Since BPMN supports many features and has an extensive syntax, this thesis aims to provide a simpler abstraction for designing privacy-aware business process models. The objective of this thesis is to design a domain-specific language (DSL) for business process models and to implement a model transformation that automatically produces a standard BPMN model from it.

The DSL should allow designers to describe the basic concepts of a business process concisely while incorporating privacy concepts into the design.

3 Tasks

1. Study the approach proposed in the work of Basin et al. [1].
2. Study the BPMN specification and review existing textual notations for BPMN. Identify the subset of BPMN to be supported by the transformation.
3. Design the syntax of the DSL and the model transformation from the DSL to the selected subset of BPMN.

4. Implement the DSL and the model transformation.
5. Validate the approach on a case study.
6. Document the design, implementation, and evaluation in the final report.

4 Deliverables

The following deliverables are due at the end of the project:

Final report Please have a look at the submission requirements in the following section. The final report should consist of an introduction, a (theoretical) background section, sections describing the work packages and their outcomes, and a conclusion. The report must be written in English.

Source code Complete, documented source code with a reproducible evaluation.

Presentation At the end of the thesis, a 20-minute presentation must be given in a thesis defense seminar. It should give an overview and discuss the most important highlights of the work.

References

- [1] David A. Basin, Søren Debois, and Thomas T. Hildebrandt. On purpose and by necessity: Compliance under the GDPR. In Sarah Meiklejohn and Kazuo Sako, editors, *Financial Cryptography and Data Security - 22nd International Conference, FC 2018, Nieuwpoort, Curaçao, February 26 - March 2, 2018, Revised Selected Papers*, volume 10957 of *Lecture Notes in Computer Science*, pages 20–37. Springer, 2018.
- [2] Object Management Group. Business process model and notation (BPMN), version 2.0, 2011. <https://www.omg.org/spec/BPMN/2.0/>.